



Fig. 5: Pilot samples from Biomonitoring 2.0 project (Image courtesy: biomonitoring2.org)



Fig. 6: Hopkins Biotech Network focuses on critical issues in biotechnology and seeks to bridge the gap between industry, academia and government (Image courtesy: hopkinsbio.org)

table data collection projects. Latest genome-sequencing projects are producing a huge volume of data linked to the function and structure of biological molecules and sequences.

Other complementary high-throughput technologies such as DNA micro-arrays are swiftly generating big amounts of data that are too overpowering for traditional methods to analyse biological data. Understanding of this rich data could deeply impact our interpretation of life at the molecular level. However, illustration of biological knowledge is a very daunting job and demands more potent and refined computational tools.

AI and heuristic methods (in particular, machine learning and data mining, cluster analysis, pattern recognition and knowledge representation) could possibly offer key solutions for the fresh hurdles posed by the progressive transformation of biology into a data-massive science. Some key areas where AI approaches are specifically encouraging and turning out to be fruitful are prediction of protein's structure and function,

semiautomatic drug design, interpretation of nucleotide sequences and knowledge acquisition from genetic data.

There is no doubt that application of AI to computational molecular biology demands exceedingly interdisciplinary and complementary skills, and these are seldom provided for in most current academic curricula. Interdisciplinary is interesting, however, unless AI and computational biology communities collaborate closely, development of new methodologies and algorithms is likely to lose pace.

Applying GIS to bioinformatics

Generally, one would not imagine geographic information systems (GIS) and bioinformatics to have any connection as these are perceived to be two different areas. However, digital maps, managing, accessing and analysing large amounts of data and research involving visualisation are some common areas.

Researchers apply GIS techniques and tools to find and track bulky patterns such as geographic distribution of cancer and other diseases in human, animal and plant populations.

Researchers in bioinformatics end up tracking very small patterns such as those in a DNA structure that might predispose an organism to develop cancer. Until recently, GIS and spatial epidemiology lay emphasis on finding and monitoring large-scale, population based occurrences such as disease clusters, outbreak of infection and possible associations between cancers and environmental factors.

Now, researchers are establishing a connection between individual-level processes tracked by genomics, and proteomics and population-level outcomes tracked by GIS and epidemiology. This will result in monitoring, quantifying and predicting human-health costs associated with the environment more effectively. In turn, associated fields, for example, climate change, emerging and resurgent infectious diseases and

environmental health will be widely benefitted.

Merging bioinformatics with GIS can be very useful in forecasting public health outcomes. Genomics and proteomics data, along with GIS, could help epidemiologists study the outbreak of diseases, how the incidence of congenital disease varies over space, over time and with changes to terrain and climate, apart from other aspects.

Future opportunities

Presently, bioinformatics is steered by a specialised group of individuals, such as database curators, who are generally doctoral-level biologists, database and software engineers, and computational biologists. However, bulk of the biologists, who are the users of the very basic bioinformatical tools, fall on the other extreme.

This is a big constraint in the bioinformatics sector today. In times to come, it is expected that majority of biologists will have possessed some basic skills such as programming, database development and management of large datasets. Hence, biologists are likely to spend more time on the computer than in their labs.

Heterogeneity of data analysis and lack of connectivity among available data are some other issues that are future development areas.

Biotech, as well as pharmaceutical firms, requires extensive software support in terms of writing algorithms, developing software for existing algorithms, managing databases and final processing of drug discovery. The large pharmaceutical and genome based biotech companies across the globe are high-potential clients for Indian IT companies to deliver complete database solutions.

The main opportunity areas for IT companies in India are improving content and utility of databases, developing and launching superior tools for data generation, and capturing, postscripting, developing and refining tools and databases for comprehensive functional studies. ●